
EXERCISES

Lecture 2: Recap

AI508 — Algorithmic Game Theory

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Contents

1	Probability	3
1.1	Random variables	4
1.2	Conditional probability	6
2	Conditional expectation	7
3	Inequalities	8

1 Probability

Serves to understand outcomes with some probability

Definition: Sample space

The space which an event can hold.

Example: Different sample spaces

- Coins $\Omega = \{H, T\}$
- Die $\Omega = \{1, 2, 3, 4, 5, 6\}$
- Real values $\Omega = \mathbb{R}$

Definition: Event

An Event is a subset of the sampling space.

Example: Different events

Given that A is an event in Ω such that $0 \leq P(A) \leq 1$, then $P(\Omega) = 1$ and $P(\phi) = 0$

Remember that

$$P(A^C) = 1 - P(A)$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

1.1 Random variables

Definition: Random Variable

The random variable X is a function that maps

$$\mathcal{X} : \Omega \rightarrow \mathbb{R}$$

$$\mathbb{E}[X] = \sum_{x \in \mathcal{X}} x \cdot P(\mathcal{X} = x)$$

Example: Coin

Two coins giving heads

$$\mathcal{X} = \{0, 1, 2\}$$

Example: Discrete random variable

Given that $\Omega = [0, 1]$, then

$$X(\omega) = \begin{cases} -1 & w \leq \frac{1}{2} \\ 1 & w > \frac{1}{2} \end{cases}$$

Definition: Probability mass function

Given $\mathcal{X}$

$$P(\mathcal{X} = x) = p_x$$

Definition: Joined distribution

Given $\mathcal{X}, \mathcal{Y}$

$$P(\mathcal{X} = x, \mathcal{Y} = y) = p_{xy}$$

Definition: Marginalization

Given $\mathcal{X}, \mathcal{Y}$

$$P(\mathcal{X} = x) = \sum_{y \in \mathcal{Y}} P(\mathcal{X} = x, \mathcal{Y} = y)$$

Definition: Bernoulli

Given that $\mathcal{X} = \begin{Bmatrix} 1 & p \\ 0 & 1-p \end{Bmatrix}$

So that $\mathbb{E}[\mathcal{X}] = 1p + 0(1-p) = p$

Definition: Variance

Variance captures the spread of value around the

$$\text{Var}[\mathcal{X}] = \mathbb{E}[(\mathcal{X} - \mathbb{E}[\mathcal{X}])^2]$$

Example: Variance on bernoulli

Given X ,

$$\text{Var}[\mathcal{X}] = \sum_{x \in \mathcal{X}} (x - \mu)^2 P(\mathcal{X} = x)$$

Definition: Covariance

Given $\mathcal{X}, \mathcal{Y}$,

$$\begin{aligned} \text{Cov}[\mathcal{X}, \mathcal{Y}] &= \mathbb{E}[(\mathcal{X} - \mathbb{E}[\mathcal{X}])(\mathcal{Y} - \mathbb{E}[\mathcal{Y}])] \\ &= \mathbb{E}[\mathcal{X}\mathcal{Y} + \mathbb{E}[\mathcal{X}]\mathbb{E}[\mathcal{Y}] - \mathcal{X}\mathbb{E}[\mathcal{Y}] - \mathcal{Y}\mathbb{E}[\mathcal{X}]] \\ &= \mathbb{E}[\mathcal{X}\mathcal{Y}] + \mathbb{E}[\mathcal{X}]\mathbb{E}[\mathcal{Y}] - \mathbb{E}[\mathcal{X}]\mathbb{E}[\mathcal{Y}] - \mathbb{E}[\mathcal{Y}]\mathbb{E}[\mathcal{X}] \\ &= \mathbb{E}[\mathcal{X}\mathcal{Y}] - \mathbb{E}[\mathcal{X}]\mathbb{E}[\mathcal{Y}] = \text{Cov}[\mathcal{X}] \end{aligned}$$

1.2 Conditional probability

Assume an implicit sample space.

Definition: Conditional probability

Assume there are 2 events A and B and let $P(B) > 0$

Then the probability of event A given B is:

$$P(A | B) = \frac{P(A \cap B)}{P(B)}$$

Example: Conditional probability

Given $B = \{x > 3\} = \{4, 5, 6\}$

The probability that $A = \text{EVEN} = \{2, 4, 6\}$ is:

$$\frac{P(A \cap B)}{P(B)} = \frac{\frac{2}{6}}{\frac{3}{6}} = \frac{2}{3}$$

Definition: Bayes Rule

Given A, B

$$P(A | B) = \frac{P(B | A)P(A)}{P(B)}$$

Where $P(B | A)$ is the *likelihood*, $P(A)$ is the *prior* and $P(B)$ is the *marginal evidence*

Definition: Independent events of a random variable

Event A and B are independent if and only if

$$P(A \cap B) = P(A)P(B)$$

Example: Given that $B > 0$

Given that $B > 0$ then

$$P(A | B) = \frac{P(B | A)P(A)}{P(B)} = \frac{P(A)P(B)}{P(B)} = P(A)$$

Definition: Independence of 2 random variables

2 R.V's X, Y are independent iff

$$P(\mathcal{X} = x, \mathcal{Y} = y) = P(\mathcal{X} = x)P(\mathcal{Y} = y) \quad \forall x, y \in (x \in \mathcal{X}, y \in \mathcal{Y})$$

Definition: Uncorrelation of 2 random variables

2 R.V's are uncorrelated iff

$$\text{Cov}(\mathcal{X}, \mathcal{Y}) = 0 \implies \mathbb{E}[\mathcal{X}\mathcal{Y}] = \mathbb{E}[\mathcal{X}]\mathbb{E}[\mathcal{Y}]$$

Example: Independence and Uncorrelation

Note that

$$\text{Independence} \implies \text{Uncorrelation}$$

2 Conditional expectation

Note that conditional expectations are not scalars,

Definition: Conditional expectation

Given $\mathcal{Y}$

$$\mathbb{E}[\mathcal{X} \mid \mathcal{Y} = y] = \sum_{x \in \mathcal{X}} x \cdot P(\mathcal{X} = x \mid \mathcal{Y} = y)$$

Theorem: Law of Total Expectation

Given X and Y ,

$$\begin{aligned} \mathbb{E}[\mathcal{X}] &= \mathbb{E}[\mathbb{E}[\mathcal{X} \mid \mathcal{Y} = y]] \\ &= \sum_{y \in \mathcal{Y}} \left(\sum_{x \in \mathcal{X}} x \cdot P(\mathcal{X} = x \mid \mathcal{Y} = y) \right) P(\mathcal{Y} = y) \end{aligned}$$

3 Inequalities

Definition: Markov's inequality

Let $\mathcal{X}$ be a non negative random variable, then.

$$P(X \leq a) \leq \frac{\mathbb{E}[\mathcal{X}]}{a}$$

Proof of Markov's inequality

Let

$$\begin{aligned}\mathbb{E}[\mathcal{X}] &= \sum_{x \geq 0} x \cdot P(\mathcal{X} = x) \\ &= \sum_{x=0}^a x \cdot P(\mathcal{X} = x) + \sum_{x \geq a} x \cdot P(\mathcal{X} = x) \\ &= \sum_{x \geq a} x \cdot P(\mathcal{X} = x) \geq a \sum_{x \geq a} P(\mathcal{X} = x) = a\end{aligned}$$

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